

Recent Advances in Cross-linked Polyethylene Based Nanocomposites for High Voltage Engineering Applications: A Critical Review

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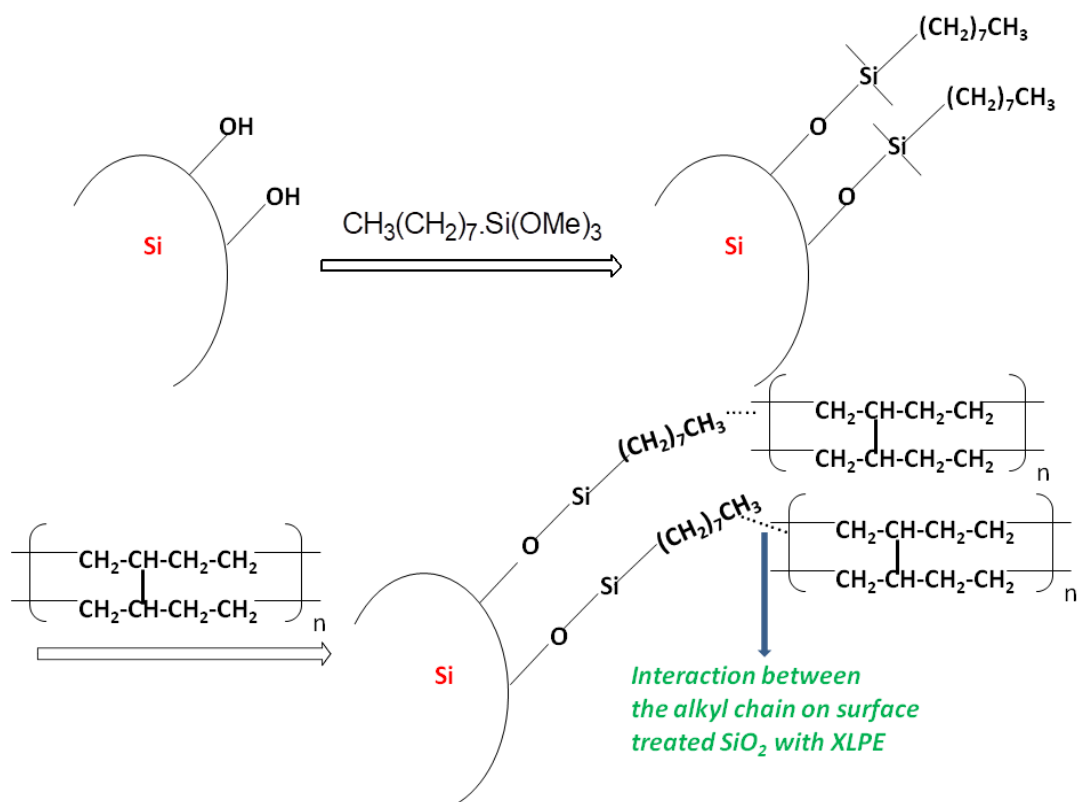


Figure S1: Interaction between surface modified SiO₂ and XLPE chain(Adapted with permission from ref.¹ Copyright 2014, The Royal Society of Chemistry.)

Table S1. Advantages and disadvantages of materials used in high voltage cable insulation fields.

Materials used	Advantages	Limitations	Reference
Paper-Insulated Lead-Covered cables (PILC)	Long-term reliability and their ability to fit it into older conduit-manhole systems.	Corrosion of the lead sheath on PILC cables.	2, 3
Polyvinyl chloride (PVC)	Cheap, Durable, widely available	Highest dielectric losses, melts at high temperatures, contains halogens.	4, 5
Polyethylene (PE)	Lowest dielectric losses, high initial dielectric strength	Highly sensitive to water treeing, material breaks down at high temperatures	6, 7
Ethylene- Propylene Rubber (EPR)	Increased flexibility, low sensitivity to water treeing	Medium-high dielectric losses	8, 9
Crosslinked Polyethylene (XLPE)	Higher operational temperature, mechanically robust and weightless,	Space charge accumulation and stress inversion phenomena	10, 11

	installation process is simpler and faster		
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Table S2. A brief insulation property comparison of XLPE with other insulation materials

Properties	Insulation Materials				
	PVC	EPR	Impregnated paper	LDPE	XLPE
Permittivity	8	3.3	3.5	2.3	2.5
Dielectric loss tangent (10 ⁻⁴)	1000	40	20	4	4
Thermal resistivity (°CmW ⁻¹)	6	5	5-6	3	3.5

Table S3. The failure mechanism and their effects in cable insulation systems

Failure factor	Mechanism	Effects
Thermal		
High temperature	Chemical reaction, incompatibility of materials, thermal expansion (radial and axial), diffusion.	Hardening, softening, loss of mechanical strength, embrittlement, increased tan delta, shrinkage, loss of adhesion, separation, delamination at interfaces, conductor penetration, formation of soft spots, wrinkles.
Low temperature	Cracking, thermal contraction.	Shrinkage, loss of adhesion, separation, delamination at interfaces.
Electrical		
Voltage, AC, DC, impulse	Partial discharges (PD), electrical treeing (ET) water treeing (WT), dielectric losses and capacitance charge injection, intrinsic breakdown	PD and ET increased, thermal ageing, immediate failure
Mechanical		
Tensile, compressive, shear stresses, Fatigue, cyclic	Cracking, rupture	Mechanical rupture, loss of adhesion, separation,

bending, vibration		delamination at interfaces
Environmental		
Water/humidity Liquids/gases contamination	Dielectric losses and capacitance, electrical tracking, water treeing, corrosion	Increased temperature, thermal ageing, thermal runaway, increased losses and ET

Table S 4. Changes in electrical properties of XLPE nanocomposites

Property	Outcome	Reason for change in properties
Permittivity	Increase/Decrease	Different permittivity values and weight percentage of nanoparticles
Dielectric loss	Well within the range of Insulators	Interface structures between XLPE and nanofiller
Partial discharge resistance	Increase	Nano segmentation of XLPE, shape of the nanofiller and electric field at interfacial region
Treeing resistance	Increase	Nano segmentation of XLPE, shape of the nanofiller and electric field at interfacial region
Breakdown strength	Increase or no change	Coupling, impurity effects and loading weight percentage of nanoparticles
DC conductivity	Increase or no change	Roles of ion traps and impurities

Table S 5. Electrical, mechanical and thermal properties of XLPE nanocomposites

PE based nanocomposites	Properties		
	Electrical (DC breakdown strength) kV/ mm	Mechanical (Tensile strength) MPa	Thermal stability (°C)
XLPE+SiO ₂ (surface modified)	446 (25°C),422 (60°C),220(80°C) ¹²	10.3 ¹³	Stable up to 540 ¹³
XLPE+ MgO	57 ¹⁴	12 ¹⁵	550 ¹⁵
LDPE +MgO	28 (AC breakdown strength) ¹⁴	12 ¹⁶	520 ¹⁶
LDPE +SiO ₂	7(AC breakdown strength) ¹⁴	10.79 ¹⁷	492 ¹⁸
LDPE +MMT (5 wt%)	19(AC breakdown strength) ¹⁴	13.6 ¹⁹	480 ^{19,20}
PE+MMT (20 wt%)	20 (AC breakdown strength) ¹⁴	18.8 (Yield strength) ²¹	455 ²¹

Table S 6. Lists of nanoparticles used in XLPE system

Nanoparticles	Findings	Reference
CaCO ₃	Improvement in the electrical performance of cross-linked polyethylene high voltage cables.	22,23
Carbon black	Decrease of dielectric constant with frequency.	24
Graphene Oxide	Improvements in thermal stability and fire safety properties in nonpolar polymers.	25,26
ZnO	Enhanced DC dielectric breakdown strength.	27,28
TiO ₂	Space charge accumulation was suppressed by filling TiO ₂ nanoparticles	29

Table S 7. Young's modulus of XLPE (X), XLPE/Al₂O₃ (A), XLPE/clay (C), XLPE/Al₂O₃ : clay = 1 : 1 (A1C), XLPE/Al₂O₃ : clay = 2 : 1 (A2C) (Reprinted with permission from ref.¹. Copyright 2014, The Royal Society of Chemistry.)

Sample	Young's modulus (MPa)
X	263± 10
C	370 ± 8
A	384 ± 6
A1C	812 ± 9
A2C	555 ± 11

References

- (1) Jose, J. P.; Thomas, S., Alumina–clay Nanoscale Hybrid Filler Assembling in Cross-Linked Polyethylene Based Nanocomposites: Mechanics and Thermal Properties. *Phys. Chem. Chem. Phys.***2014**, *16*, 14730.
- (2) Dyba, J., The Rise and Decline in The United States of Impregnated Paper Insulated Metallic Sheathed Cable, Solid-type. I. *IEEE Electr. Insul. Mag.* **1999**, *15*, 13.
- (3) Goodwin, F.; Dyba, J. In Recent Technological Developments for Lead Sheathed Cables, Proceedings of *IEEE/PES Transmission and Distribution Conference*, IEEE; **1994**, 568-573.
- (4) Beneš, M.; Plaek, V.; Matuschek, G.; Kettrup, A.; Györyová, K.; Emmerich, W.; Balek, V. Lifetime Simulation and Thermal Characterization of PVC Cable Insulation Materials. *J. Therm. Anal. Calorim.* **2005**, *82*, 761.
- (5) Yaacob, M.; Sin, L.; Aman, A. In A New Polyvinyl Chloride Cable Insulation Using Micro and Nano Filler Materials, 4th *International Power Engineering and Optimization Conference* (PEOCO), IEEE; **2010**, 221-225.
- (6) Vahlstrom, W., Investigation of Insulation Deterioration in 15 kV and 22 kV Polyethylene Cables Removed from Service. *IEEE Trans. Power Appar. Sys.* **1972**, *3*, 1023.
- (7) Serra, S.; Montanari, G.; Mazzanti, G. Theory of Inception Mechanism and Growth of Defect-induced Damage in Polyethylene Cable Insulation. *J. Appl. Phys.***2005**, *98*, 034102.
- (8) Brown, M. Performance of Ethylene-propylene Rubber Insulation in Medium and High Voltage Power Cable. *IEEE Trans. Power Appar. Sys.* **1983**, *2*, 373.
- (9) Mizuno, Y.; Nagao, M.; Kosaki, M.; Shimizu, N.; Horii, K. Evaluation of Ethylene-Propylene Rubber as Electrical Insulating Material for A Superconducting Cable. *IEEE Trans. Electr. Insul.***1992**, *27*, 1108.
- (10) Chen, G.; Hao, M.; Xu, Z.; Vaughan, A.; Cao, J.; Wang, H. Review of High Voltage Direct Current Cables. *CSEE J. Power Energy Sys.* **2015**, *1*, 9.

- (11) Fabiani, D.; Montanari, G. C.; Laurent, C.; Teyssedre, G.; Morshuis, P.; Bodega, R.; Dissado, L.; Campus, A.; Nilsson, U. Polymeric HVDC Cable Design and Space Charge Accumulation. Part 1: Insulation/Semicon Interface. *IEEE Electr. Insul. Mag.* **2007**, 23, 11.
- (12) Roy, M.; Nelson, J. K.; MacCrone, R.; Schadler, L. Candidate Mechanisms Controlling the Electrical Characteristics of Silica/XLPE Nanodielectrics. *J. Mater. Sci.* **2007**, 42, 3789.
- (13) Jose, J. P.; Ahmad, Z.; Thomas, S., Hybrid Nanoparticle-Based XLPE/SiO₂/TiO₂ and XLPE/SiO₂ Nanocomposites: Nanoscale Hybrid Assembling, Mechanics and Thermal properties. In *InCIEC* Springer; **2014**, 895-902.
- (14) Paramane, A. S.; Kumar, K. S., A review on Nanocomposite Based Electrical Insulations. *Trans. Electr. Electron. Mater.* **2016**, 17, 239.
- (15) Peng, S.; He, J.; Hu, J.; Huang, X.; Jiang, P. Influence of Functionalized MgO Nanoparticles on Electrical Properties of Polyethylene Nanocomposites. *IEEE Trans. Dielec. Electr. Insul.* **2015**, 22, 1512.
- (16) Reddy, C. C.; Gosyowaki, M.; Murata, Y.; Sekiguchi, Y. In Superior Thermal Breakdown Performance of MgO-LDPE Nanocomposite Materials for HVDC Insulation. International Symposium on *Electrical Insulating Materials (ISEIM 2008)*, IEEE; **2008**, 661-664.
- (17) Redhwi, H. H.; Siddiqui, M. N.; Andrady, A. L.; Hussain, S. Durability of LDPE Nanocomposites with Clay, Silica, and Zinc oxide: Part I: Mechanical Properties of the Nanocomposite Materials. *J. Nanomaterials*. **2013**, 21.
- (18) Wang, Y.; Wang, C.; Zhang, Z.; Xiao, K. Effect of Nanoparticles on the Morphology, Thermal, and Electrical Properties of Low-density Polyethylene After Thermal Aging. *Nanomaterials*, **2017**, 7, 320.
- (19) Monica A, P.; Bernabé L, R.; Karla A, G.-M.; Víctor H, C.-R.; Miguel, M.; Johanna, C.; Álvaro, M., Low Density Polyethylene (LDPE) Nanocomposites with Passive and Active Barrier Properties. *J. Chile. Chem. Soc.* **2014**, 59, 2442.
- (20) Liu, S.-P. Flame Retardant and Mechanical Properties of Polyethylene/Magnesium hydroxide/Montmorillonite Nanocomposites. *J. Ind. Eng. Chem.* **2014**, 20, 2401.
- (21) Araújo, E.; Barbosa, R.; Rodrigues, A.; Melo, T.; Ito, E. Processing and Characterization of Polyethylene/Brazilian Clay Nanocomposites. *Materials Science and Engineering: A*. **2007**, 445, 141.

- (22) Habib, M.; Nasart, L.; Sharkawy, R. In Improvement The Electrical Performance of Cross-linked Polyethylene High Voltage Cables, *Power Systems Conference (MEPCON)*, Eighteenth International Middle East, IEEE; **2016**; 21-25.
- (23) Ding, J.; Ma, W.; Song, F.; Zhong, Q. Effect of Nano-calcium Carbonate on Microcellular Foaming of Polypropylene. *J.Mater.Sci.***2013**, *48*, 2504.
- (24) Moon, K. S.; Do Choi, H.; Jung, H. Y.; Cho, K. Y.; Yoon, H. G.; Moon, T. J. Dielectric Properties of Human Phantom Material Using XLPE Composite With Shape Memory Characteristics. *Polymer (Korea)*.**1999**, *23*, 763.
- (25) Hu, W.; Zhan, J.; Wang, X.; Hong, N.; Wang, B.; Song, L.; Stec, A. A.; Hull, T. R.; Wang, J.; Hu, Y. Effect of Functionalized Grapheneoxide With Hyper-branched Flame Retardant on Flammability and Thermal Stability of Cross-linked Polyethylene. *Ind.Eng.Chem. Res.***2014**, *53*, 3073.
- (26) Tripathi, S. N.; Saini, P.; Gupta, D.; Choudhary, V. Electrical and Mechanical Properties of PMMA/reduced Graphene oxide Nanocomposites Prepared via in situ Polymerization. *J. Mater Sci.* **2013**, *48*, 6223-6232.
- (27) Kim, Y.-M.; Cha, Y.-K.; Lim, K.-J.; Nam, J.-H.; Lee, G.-J. In Electrical Insulation Evaluation of Crosslinked Polyethylene Nanocomposite Blended with ZnO, International Conference on *Condition Monitoring and Diagnosis (CMD)*, IEEE; **2012**, 1242-1245.
- (28) Hong, J.; Schadler, L.; Siegel, R.; Mårtensson, E. Electrical Behavior of Low Density Polyethylene Containing an Inhomogeneous Distribution of ZnO nanoparticles. *J.Mater. Sci.* **2006**, *41*, 5810.
- (29) Wang, Y.; Xiao, K.; Wang, C.; Yang, L.; Wang, F. Effect of Nanoparticle Surface Modification and Filling Concentration on Space Charge Characteristics in TiO₂/XLPE nanocomposites. *J. Nanomaterials*. **2016**.